



Designing meaningful human oversight in AI

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Abstract

Human oversight is central to safe and responsible AI, but current approaches risk either collapsing agentic AI into mere automation, stripping it of its agentic character, or reducing human agency to a rubber stamp. This paper proposes a design framework that treats agency as layered: AI operative agency in task execution, and human evaluative agency in verification, steering, and substitution. Instead of demanding low-level explanations and controls over how a complex AI model works internally (i.e. internal reasoning faithfulness), we focus on high-level explanations tied to external criteria and human expert understanding (external reasoning faithfulness). This approach retains AI's operative agency while strengthening human's evaluative agency. We also exploit the solve-verify asymmetry by designing AI outputs so that humans can efficiently check and contest them without having to resolve the task. This paper makes three contributions. First, it develops a layered agency framework that distinguishes operative and evaluative agency and specifies where human accountability attaches in AI-enabled decision systems. Second, it reframes the explainability requirement by arguing that external reasoning faithfulness—alignment with externally articulated criteria and human expertise—is sufficient and often preferable to internal mechanistic transparency for enabling meaningful oversight. Third, it provides a structured catalogue of oversight mechanisms (e.g., structured rationales, reasoning traces, confidence signals, policy attribution, circuit breakers, appeal bundles) and four end-to-end design patterns that translate these principles into implementable system architectures. We also outline evaluation criteria for AI's agency, human's agency, and joint system agency. The framework provides AI ethicists, engineers, safety teams, users, and organisational leaders with a concrete way to design meaningful and effective oversight that preserves human accountability and agency while allowing AI to retain its agentic features and autonomy.

Keywords Artificial intelligence · AI · AI agency · Human oversight · Human agency

1 Introduction

Powerful AI and agentic systems, especially generative models often treated as black boxes due to limited explainability, are increasingly deployed in critical domains and high-risk use cases where their outputs shape individual lives, organisational outcomes, and public policy. Calls for “meaningful human oversight” are now common across technical, ethical, and legal frameworks, from industry safety standards to the European Union's AI Act. By oversight we mean the structured capacity of humans within a socio-technical system to understand, evaluate, and where necessary intervene in or override the outputs of a computational system, such that accountability for consequential decisions remains with identifiable human agents. Yet what counts as meaningful and effective oversight remains under-defined and is often reduced to a checklist of the mere presence of different

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oversight measures [1]. In practice, oversight often fails in one of two ways. Either humans act as rubber stamps, approving AI outputs they do not fully understand, or systems are constrained so tightly that AI collapses into rule-following automation, losing the very agentic qualities that make AI highly useful [2].

This paper begins from the observation that human agency and AI agency are not a zero-sum game. By agency we mean a system's genuine capacity to steer outcomes toward a goal [3]. Adding human involvement does not necessarily strip agency from the AI. The key is to distinguish **operative agency**, the AI's capacity to generate solutions, from **evaluative agency**, the human's capacity to understand, judge, verify the AI's solution, and, where necessary, steer its outcomes or substitute them with alternative solutions. We use the term evaluative rather than oversight to emphasise that the human role is not merely supervisory or symbolic but actively involves assessment and contextual judgement. For example, an AI system exercising operative agency may generate a highly accurate diagnosis or treatment plan by processing imaging, lab results, and the patient's full medical history. The human doctor's role, grounded in evaluative agency, is not to replicate this diagnostic process nor to surpass the AI's technical accuracy. Instead, it is to understand and evaluate whether the AI's rationale is consistent with what a responsible human expert would find explainable and justifiable, and to judge its appropriateness in light of nuanced context and value-laden considerations, such as treatment burden, patient preferences, or ethical standards. In this way, evaluative agency provides accountability and higher-order judgement that anchor the AI's operative agency within acceptable professional and societal boundaries.

Excessive control and intervention powers without understanding and evaluative agency are of little value: if humans lack the ability to make sense of when and why to intervene, their role reduces either to symbolic oversight, signing off without adding substance, or to blunt overwriting of AI outputs, which risks discarding correct decisions and degrading overall performance. These concerns are central to philosophical and legal accounts of Meaningful Human Control (MHC). These accounts argue that effective control over AI systems cannot be reduced to moment-to-moment human intervention, but must be distributed across design-time constraints, accountability structures, and run-time decision points. This paper adopts MHC as a conceptual starting point and asks how such control can be operationalised in practice. We answer this by reframing human oversight as an allocation of agency between AI systems and human evaluators, distinguishing operative agency from evaluative agency. Designed well, these two forms of agency can coexist: AI systems retain their operative agency in “doing”

the work, while humans exercise evaluative agency through improved understanding that preserves both accountability and effectiveness.

The main contributions of this paper are: (1) a conceptual framework for oversight grounded in agency, distinguishing operative agency (the AI's capacity to generate solutions) from evaluative agency (the human's capacity to assess, contest, and override them), which addresses the responsibility gap in AI ethics by specifying where human accountability attaches; (2) a reframing of the explainability requirement, arguing that external reasoning faithfulness is sufficient and often preferable to internal mechanistic transparency for enabling meaningful oversight, a position that engages with the normative vs. epistemic explainability debate; and (3) a catalogue of oversight mechanisms and end-to-end design patterns that engineers, policymakers, and AI ethics researchers can adapt across contexts.

The remainder of the paper is organised as follows. Section 2 discusses the related work. We then present Sect. 3 to introduce the conceptual foundations. The methodology of this study is introduced in Sect. 4. Section 5 discusses goals and principles for designing oversight. Section 6 presents a catalogue of oversight mechanisms and Sect. 7 introduces four design patterns. Section 8 discusses the limitations and challenges of the study and Sect. 9 concludes the paper.

2 Related work

A wide range of AI ethics guidelines and emerging regulations emphasise the need for meaningful human oversight of automated systems. For example, the EU's High-Level Expert Group on AI [4] identified “human agency and oversight” as a key requirement for trustworthy AI, and the proposed EU AI Act [5] explicitly mandates that high-risk AI systems be designed to allow effective human oversight. ISO/IEC 42001:2023 [6], the AI management system standard, emphasises oversight as governance routines and roles. The Australian Mandatory Guardrails for AI [7] also includes human oversight as a mandatory guardrail in high-risk settings, highlighting the need to minimise risks during the deployment and operation. While these regulations and standards articulate oversight at a high level, they provide limited actionable guidance on the concrete mechanisms and interaction contracts at the AI-human interface.

In academia, several studies focus on human oversight. Green [8] finds that oversight provisions in dozens of algorithmic governance policies often assume humans can reliably correct AI errors, an assumption undermined by evidence of human limitations. Studies have shown that humans sometimes reject correct AI outputs due to

low trust or an inflated confidence in their own judgment, thereby nullifying potential benefits of the AI [9]. Parasuraman and Manzey [10], for instance, describe how operators may either become complacent monitors who miss critical alerts, or conversely, intervene too often and “turn off” the automation even when it is performing well. Similarly, Green and Chen [11] observed that human decision-makers sometimes override accurate algorithmic recommendations, degrading overall decision quality. Hasan [12] proposes the MUSE framework, which integrates multimodal representation, generative simulation, and symbolic grounding to improve human-centered modeling of experience. It emphasises the creation of interpretable rationale–evidence artefacts, although it was developed independently and within a different domain. MUSE focuses on understanding at the individual experiential level, while our framework applies similar principles of contextual and symbolic reasoning to the oversight relationship between humans and AI agents.

Beyond generic “human-in-the-loop”, philosophical accounts of Meaningful Human Control (MHC) argue for reason-responsive control distributed across design-time and run-time structures, such as who takes responsibility, what levers exist, and how systems track human values [13]. This complements regulatory oversight by clarifying the locus of control and responsibility in socio-technical systems, reinforcing our agency-first design stance. Unlike traditional human-in-the-loop models, which treat human involvement as a control checkpoint, our framework aligns more closely with human-AI teaming research. In particular, it reflects principles of adjustable autonomy [14, 15] and mixed-initiative systems [16, 17], where responsibility dynamically shifts based on task difficulty, uncertainty, and impact. Operative agency corresponds to AI-led initiative in solution generation, while evaluative agency captures human-led initiative in judgment, verification, and goal alignment.

Human-AI interaction (HAI) research examines what improves team performance in collaboration. As “explanations” alone rarely fix inappropriate reliance, interventions that change the interaction (e.g., who decides, when, and with what friction) tend to help more [9, 10]. Bućinca et al. [18] show that cognitive forcing functions (e.g., requiring an initial judgment before seeing AI) reduce over-reliance. Some studies report mixed or negative effects of generic transparency on users’ ability to catch AI errors, reinforcing the need for task-tuned oversight affordances rather than one-size-fits-all Explainable AI (XAI). Much of the XAI literature focuses on explaining model internals or local input–output sensitivities, including saliency maps, feature attributions, and counterfactual explanations [19, 20]. While valuable for debugging and model development, these approaches often fail to support effective human

oversight in high-stakes operational settings, where humans must judge outcomes against external criteria such as policy, law, or professional norms. Our notion of external reasoning faithfulness differs from traditional XAI by shifting the target of explanation away from model internals and toward externally grounded justification that supports verification, contestation, and accountability. HAI design guidelines [21] consolidate such patterns into actionable rules (set expectations, support contestation, expose uncertainty, and adapt over time), which map well to our mechanisms catalogue.

Beyond governance and HAI research, the framework also connects to several live debates in AI ethics. First, it responds to the responsibility gap identified by Matthias [22]: as AI systems grow more autonomous, neither the developer nor the user may be clearly accountable for outcomes. Our operative–evaluative distinction addresses this gap directly by specifying where human accountability attaches, not to replicating or co-owning the AI’s generative process, but to evaluating its outputs against normative criteria. Second, the framework engages with contestability as a democratic value [23]: the appeal bundle mechanism and structured rationale schema are designed to make AI decisions not merely transparent but practically challengeable. Third, our treatment of subjective tasks in Sec. 3.4 reflects the value pluralism concern raised by Floridi and Cowls [24]: rather than forcing a single answer, we propose that oversight surface multiple perspectives and make trade-offs explicit. Fourth, the distinction between internal and external reasoning faithfulness responds to ongoing debates about whether explainability is primarily oriented toward epistemic goals (understanding how the model works) or normative goals (enabling accountability and justified trust) [25]. We argue that for oversight purposes, the normative framing is primary.

In this study, existing regulations and standards explain why oversight is required; HAI and human-factors research suggest what tends to work; AI ethics debates clarify what values oversight must serve; and assurance research offers how to build it. We integrate these by reframing oversight responsibilities, shifting to external reasoning faithfulness, and providing a mechanisms-by-context catalogue of concrete oversight surfaces.

3 Conceptual foundations: agency, understanding, and solve-verify asymmetry

3.1 Agency

Agency is a system’s genuine capacity to steer outcomes toward a goal. Following Barandiaran et al. [26], agency can be described through four conditions: individuality (the

presence of a system boundary), source of action (causal origin of outputs), goal-directedness (pursuit of norms or objectives), and adaptivity (the ability to adjust behaviour in light of feedback). The four agency conditions are not treated as abstract descriptors but as architectural design levers guiding the structure of the framework. Individuality, understood as the presence of a system boundary, is operationalised through explicit role separation and provenance tracking, ensuring that actions and interventions are attributable to distinct actors within the decision stack. Interactional asymmetry, defined as the capacity to causally influence system trajectories, is implemented through divergence detection, escalation triggers, and override mechanisms that ensure evaluative intervention can meaningfully redirect outcomes rather than merely observe them. Goal-directedness, or normativity, is instantiated through goals ledgers, structured justification artefacts, and policy-aligned evaluation criteria that make evaluative standards explicit and contestable. Finally, adaptivity, understood as the capacity to adjust behaviour in light of feedback, is embedded through feedback loops, recalibration mechanisms, abstention protocols, and iterative review processes that allow both human and system behaviour to evolve over time. In this way, the agency conditions function as constraints on system design: they specify what must be present in the architecture for human oversight to constitute genuine agency rather than symbolic involvement.

Abel et al. [3] extend this account by showing that each condition is frame-dependent: whether an AI system is judged to have agency depends on the chosen reference frame, how boundaries, goals, causes, or adaptivity are defined. Unlike humans, whose agency is typically taken as given, an AI may or may not be considered an agent depending on the observer's framing, particularly in relation to the human's role within a socio-technical system. To clarify each condition: individuality requires that the system can be demarcated from its environment that there is an identifiable boundary separating what is 'inside' the agent from what is outside, enabling attribution of actions to a specific entity rather than to the environment at large. Source of action requires that the system is a genuine causal origin of its outputs, not merely a passive conduit for external instructions, the system contributes something that would not have occurred without its particular processing. Goal-directedness requires that the system's behaviour can be understood as pursuing objectives or norms, whether explicitly programmed or emergent from training, its outputs are not random but oriented toward identifiable ends. Adaptivity requires that the system can modify its behaviour in response to feedback or changing conditions, adjusting its approach rather than rigidly repeating fixed routines. Each of these conditions admits of degrees and, as Abel et

al. [3] emphasise, depends on the observer's frame of reference: the same system may satisfy a condition under one framing and fail it under another. For oversight design, the significance is practical. Each condition identifies a specific dimension along which the distribution of agency between human and AI must be deliberately structured.

Alternative accounts of agency in philosophy and cognitive science emphasise intentionality, free will, or moral responsibility, often focusing on metaphysical questions about whether AI systems can qualify as agents in a normative sense. In AI ethics, debates about moral agency similarly centre on attributing responsibility or rights to autonomous systems. By contrast, we adopt the frame-dependent account proposed by Barandiaran et al., which treats agency as relational and situated rather than metaphysical. This definition is particularly suited to design contexts because it specifies observable conditions under which an entity can meaningfully influence outcomes within a given frame. It therefore provides an operational basis for structuring oversight mechanisms without requiring claims about artificial moral personhood.

Building on this frame, we distinguish between two complementary forms of agency: operative agency and evaluative agency. Operative agency refers to the AI's capacity to "do the work": generating solutions in a relatively autonomous fashion. Evaluative agency refers to the human role to understand, judge, verify the AI's solution, and, where necessary, steer its outcomes or substitute them with alternative solutions. Far from being mutually exclusive, these forms of agency work in tandem: operative agency allows the AI to generate solutions without unnecessary human encumbrance, while evaluative agency provides the accountability, contestability, and normative alignment that anchor those solutions in practice. This framing responds to what Matthias [22] terms the 'responsibility gap': by anchoring human accountability in evaluative agency rather than in replicating the AI's generative process, the framework assigns clear responsibility even when the AI's internal reasoning is opaque. It also operationalises the notion of contestability as a design requirement, not merely a legal right [23].

3.2 The types of understanding

A central issue in designing oversight is what counts as "understanding." Full mechanistic insight into the internal computations of modern generative or learning models is not yet feasible [27]. Building on the established notion of reasoning faithfulness [28], we distinguish between two forms. Internal reasoning faithfulness refers to human-understandable explanations that mirror the model's low-level computations, such as algorithmic circuitry, activation patterns, or concept representations across parameters [29]. External

reasoning faithfulness, sometimes described as plausibility, refers to explanations (whether articulated by the AI or imposed afterward) that align with external criteria, established standards, and expert human judgement [30]. Due to limitations in current mechanistic interpretability, understanding via internal reasoning faithfulness is often not feasible. Even when achievable, it may reveal aspects of an AI system's inner workings that are too detailed or complex to be operationally useful for oversight. Understanding and effective oversight therefore tends to rely more on external reasoning faithfulness, which provides interpretable signals that humans can assess, contest, and hold accountable. External reasoning faithfulness should therefore be understood not as a replacement for XAI, but as a complementary oversight-oriented abstraction layer that prioritises verifiability over mechanistic transparency.

3.3 Solve-verify asymmetry

Oversight benefits from recognising the solve-verify asymmetry: solving and verifying are distinct tasks, and their relative difficulty varies by context. In many settings, solving is demanding while verification is comparatively straightforward; in others, generating a solution is trivial but verification is resource-intensive. This asymmetry matters both at design time (when systems are designed, implemented and prepared for deployment) and at runtime (when particular instances are solved and evaluated) (Table 1). Effective oversight design must exploit situations where verification is easier, while mitigating cases where verification is the bottleneck.

Design-time asymmetry

At design time, the “solve” task involves creating the system architecture, specifying objectives, implementing the system, and designing verification tasks and thresholds that will apply across many future instances. The “verify” task is the evaluation of these artefacts before deployment, through requirements review, architecture and design evaluation, testing, and other forms of verification and validation.

Table 1 Solve-Verify Asymmetry

Stage	Solve harder / Verify easier	Solve easier / Verify harder
Run-time	Hard problems (e.g., proofs, factoring, structured outputs); verification via executing proof steps, tests, rule checks	Easy generation (e.g., essays, policy drafts, translations); verification for accuracy, fidelity, nuance is costly for high-risk use cases
Design-time	Complex system design (e.g., high-assurance extraction, code assistant with tests); verification via high quality test suites	Simple system setup (e.g., chatbot endpoint, baseline recommender); verification for safety, fairness, compliance is complex

There are cases where solving is substantially harder than verifying. For example, developing a high-assurance information-extraction service requires careful schema design, integration of retrieval components, and calibration of confidence signals. Once built, however, verification can be comparatively tractable: artefacts can be stress-tested through coverage checks, constraint validation, and citation-grounding audits that demonstrate whether the system will support efficient instance-level verification later.

Conversely, there are design contexts where solving is easy but verification is difficult. Building an AI chatbot endpoint for open-ended text tasks requires relatively little design effort. Yet verifying that such a system will be safe, factual, fair, and legally compliant across domains is far more challenging, since specifications are open-ended and changing. Likewise, creating a baseline recommender system is straightforward, but verifying its long-term effects on fairness, subgroup outcomes, and feedback loops is demanding.

Runtime asymmetry

At runtime, the asymmetry plays out at the level of individual instances. Here, the “solve” task is the system exercising operative agency to generate a particular solution, while the “verify” task is a human or another system exercising evaluative agency to decide whether to accept, adjust, or contest the solution.

Many instances illustrate solving as harder than verifying. Producing a mathematical proof or factoring a large number may be computationally difficult, yet verification can be decisive by checking each step or multiplying the proposed factors. Similarly, an AI may generate structured outputs such as JSON schemas or citation lists, which can be validated automatically against predefined rules. In these situations, even hallucination could be understood as exploratory generation, acceptable so long as oversight can efficiently separate valid from invalid outputs.

Other instances reverse the asymmetry: solving is easy, but verifying is hard. A generative system can produce a long, persuasive essay or policy draft in seconds. Yet verifying its factual fidelity, completeness, and normative alignment may require hours of human expert review. In such cases, hallucinations are dangerous not because generation is inexplicable, but because verification is costly and prone to oversight gaps.

Implications for oversight

The insight is that verification should be engineered into the system itself but can be done at design time and/or runtime. When verification is generally easier and more reliable at the system level than at runtime, so as much verification as possible should be performed during design. Effective design should therefore reduce or even eliminate the need for costly runtime verification, relying instead on

lightweight monitoring, sampled verification and audit. When design-time verification is itself difficult, runtime must be supported with stronger signals, such as confidence indicators, rationale traces, and citations, so that evaluative agency has reliable cues for oversight.

3.4 Oversight of subjective tasks

In subjective domains, disagreement should be treated as a feature rather than a flaw of oversight. Rather than collapsing diverse perspectives into a single answer, effective oversight may require surfacing and preserving multiple viewpoints or counterfactuals for human evaluators to weigh in context. This reflects the value pluralism emphasised by Floridi and Cowls [24]: AI systems operating in morally contested domains should not foreclose normative disagreement but instead make trade-offs legible and open to deliberation. This approach acknowledges the inherent plurality of values and interpretations and strengthens accountability by making trade-offs explicit. At the same time, it is important to recognise the boundary of delegation, i.e., certain forms of judgment, such as legal sentencing, medical diagnosis, or hiring decisions, carry normative and ethical weight that should not be delegated entirely to AI systems. In such cases, oversight goes beyond verifying outputs and extends to safeguarding human responsibility, ensuring that AI remains a tool for support rather than substitution.

4 Methodology

The goal of this study is analytical generalisation: deriving, from these real-world cases, the recurring mechanisms and interaction patterns that inform the proposed human oversight framework.

We employed an **analysis of known uses** as the primary approach. This approach was complemented by a targeted **literature review** to provide theoretical grounding and context. These methods enable both conceptual coherence and practical plausibility in developing the human oversight framework.

As demonstrated in Fig. 1, the methodology adopted in this study consists of two main steps. The first step involved desk research to identify known uses of human oversight from publicly available sources. In the second step, we conducted targeted keyword searches to identify relevant academic works, collect supporting evidence, and refine the catalogue of mechanisms and oversight patterns. These steps were iteratively repeated to ensure the continuous refinement and validation of the mechanisms and patterns.

Known-uses analysis protocol

We conducted a structured “known-use” analysis of publicly documented AI systems to identify recurring oversight mechanisms and to test the completeness of the proposed mechanism catalogue. We searched and screened sources between June–September 2025 across:

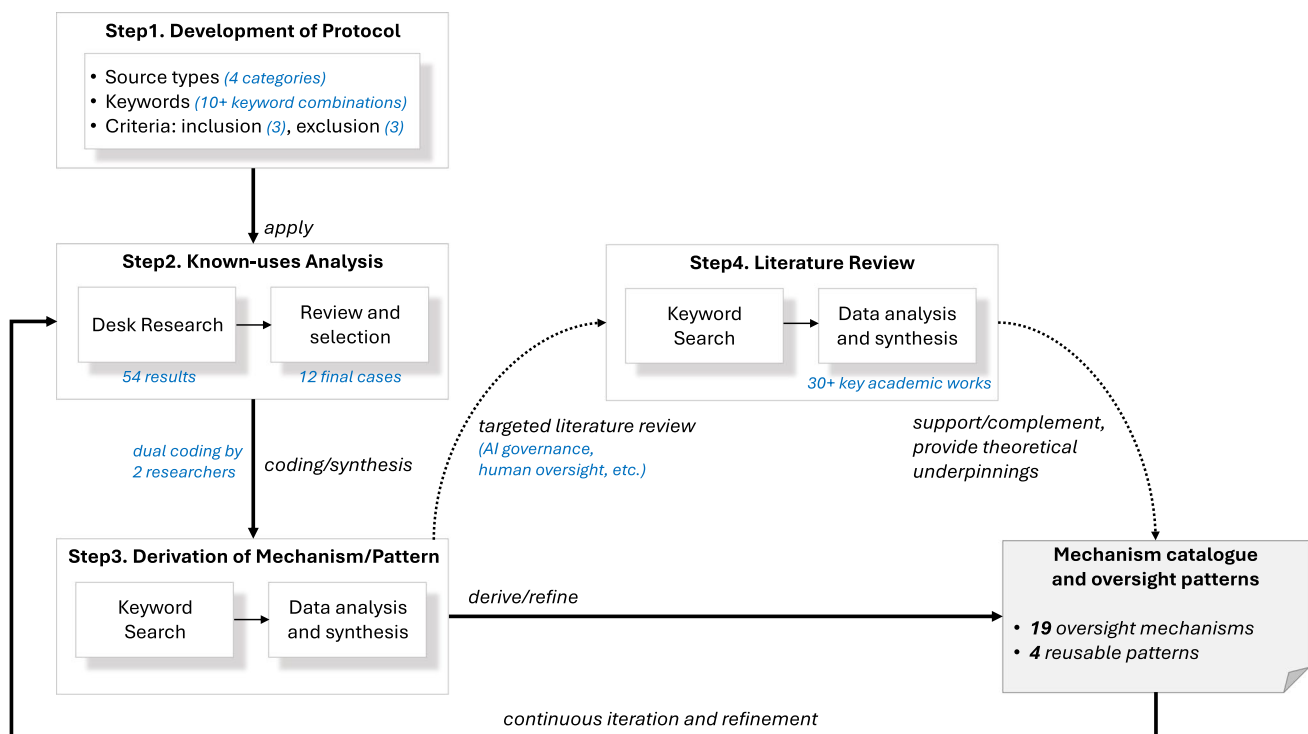


Fig. 1 Overview of the research methodology showing the primary known-uses analysis complemented by a targeted literature review to refine and synthesise oversight mechanisms and design patterns

- Vendor technical documentation and developer guides,
- Product whitepapers and architecture posts,
- Regulatory filings/impact assessments where available, and
- Independent audits or investigative reports.

Table 2 Summary of selected cases in the study

Case system	Domain	Source	Oversight mechanism observed
AAAI [31]	Academic publishing, peer review	Conference documentation and pilot reports	Human reviewer verification; structured feedback forms; final decision authority retained by editors
Revelo [32]	Recruitment, HR screening	Vendor documentation	Human-in-the-loop screening; score transparency; hiring manager override rights
Vertesia [33]	Enterprise AI workflow orchestration	Vendor technical documentation	Workflow gating; approval checkpoints; logging of prompts and outputs; configurable intervention triggers
ValidMind [34]	Financial services/model risk management	Vendor documentation and regulatory alignment materials	Audit trails; compliance checklists; documentation validation; traceability to regulatory standards
Norm AI [35]	Regulatory compliance/legal AI	Vendor documentation	Policy attribution; structured reasoning aligned to legal clauses; mandatory human review before action
Acrolinx [36]	Enterprise content governance	Vendor documentation	Rule-based scoring dashboards; configurable thresholds; human editorial override
Consensus [37]	Scientific literature search	Public documentation	Source attribution; citation linking; confidence indicators; human verification required
Elicit [38]	Research assistance, evidence synthesis	Public documentation	Structured evidence extraction; citation transparency; editable reasoning traces
Scite Assistant [39]	Scientific citation intelligence	Public documentation	Citation classification (supporting/contrasting); source traceability; confidence signals
Primer Command [40]	Security, intelligence analysis	Vendor documentation	Analyst-in-the-loop workflow; alert escalation; task assignment controls; audit logs
Dataminr [41]	Real-time event detection, crisis monitoring	Vendor documentation	Confidence scoring; analyst review layer; escalation protocols; logging mechanisms
Signal AI [42]	Media and regulatory intelligence	Vendor documentation	Source traceability; configurable alerts; human review and curation layer

To identify candidate cases, keyword queries were used, combining human oversight, review, approval, audit, traceability, appeals, provenance, policy controls, guardrails, and monitoring. The inclusion criteria for case selection is (1) an AI-enabled system deployed or trialed in a real organisational setting, (2) sufficient public detail to extract at least one concrete oversight mechanism (e.g., logging, escalation, appeals, circuit breakers), and (3) identifiable decision context (domain, decision type, users). We also defined exclusion criteria: (1) purely conceptual demos without deployment context, (2) sources that only state “human in the loop” without describing any operational mechanism, and (3) duplicate documentation for the same system/version where no new mechanism evidence was provided.

After screening 54 candidates, we retained 12 cases for analysis. For each retained system, two authors independently extracted oversight-related features and mapped them to the mechanism taxonomy; disagreements were resolved through discussion and a final adjudication pass. Table 2 presents a summary of the selected cases in this study.

We conducted a known-use analysis of these systems to identify recurring oversight mechanisms. These observations informed the development and refinement of the mechanism catalogue presented in Table 5.

Known-uses analysis

The core of this study is a structured analysis of known uses, publicly documented AI systems and oversight practices that exemplify how human oversight is implemented in practice. The aim is analytical generalisation [43] deriving recurrent mechanisms and interaction patterns from real-world systems to inform general design principles.

Known uses were identified through desk research that systematically reviewed publicly available documentation, press releases, technical white papers, and product descriptions. As presented previously, the search focused on AI systems that explicitly incorporated human oversight features, such as verification, validation, review, or audit functionalities, at either the design or operational stage. Each source was examined to determine whether oversight was integral to the system’s workflow or merely supplementary, ensuring that only cases demonstrating substantive human–AI interaction were included.

Cases were then selected to capture diversity in oversight contexts, covering a range of domains and purposes. For example,

- Academic peer review (AAAI pilot [31]),
- Compliance and documentation checking (ValidMind Document Checker [34], Norm AI [35]),
- Evidence synthesis (Consensus [37], Elicit [38], Scite [39]), and

- Real-time monitoring (Primer Command [40], Datamir [41], Signal AI [42]).

Each case was examined using document analysis [44] to extract observable features such as:

- The oversight artefacts exposed to humans (e.g., rationales, provenance, confidence signals, appeal bundles),
- The timing and level of human intervention (design-time vs run-time),
- The agency dimensions affected (individuality, source of action, goal-directedness, adaptivity), and
- How the system leverages the solve-verify asymmetry to balance AI autonomy and human evaluation.

Findings from the known-uses analysis directly informed the catalogue of oversight mechanisms (Sect. 6) and the four end-to-end oversight patterns (Sect. 7).

Derivation and scope of the four patterns

The four end-to-end patterns were derived through iterative synthesis of the mechanisms identified in the known-use analysis (Table 2) and the conceptual constraints articulated in the layered agency framework. We first clustered observed mechanisms according to their functional role in structuring the relationship between operative and evaluative agency (e.g., verification, steering, substitution, escalation). We then examined how these clusters could be composed into coherent, end-to-end configurations spanning design-time and run-time oversight.

During this process, we identified multiple candidate configurations. The four patterns presented in this paper were selected because they (i) recur across multiple domains, (ii) represent distinct allocations of operative and evaluative agency, and (iii) collectively cover the primary oversight objectives identified in the framework. They are therefore not claimed to be the only possible configurations, but rather representative and analytically distinct patterns that illustrate the design space. Additional domain-specific variants are possible and are discussed as extensions rather than separate foundational types.

Literature review

To situate these mechanisms within the broader research landscape, we conducted a **targeted literature review** [45, 46]. The review integrated work across AI governance, human-AI interaction, human oversight, and assurance engineering, including regulatory frameworks [5–7] and empirical studies on human-AI reliance and control [8, 10, 11, 13, 21]. This synthesis provided the theoretical underpinnings for key constructs such as operative vs evaluative agency, external reasoning faithfulness, and the solve-verify asymmetry, which shaped the analytical framework used in the known-uses mapping.

The combined approach provides a theory-informed analytical validation. Mechanisms and patterns were assessed against oversight design goals, including control, contestability, competence, and transparency, to test their plausibility and transferability across domains. The evaluation is interpretive rather than experimental, focusing on conceptual soundness and cross-domain consistency.

Both data sources (academic literature and public documentation) are secondary, which means they are not directly generated or collected from interviews with AI engineers, user testing of oversight tools, controlled experiments. Accordingly, results depend on the accuracy and completeness of available information. Future work will include empirical testing with developers and reviewers to measure oversight effectiveness, intervention accuracy, and verification efficiency.

5 Oversight as agency allocation: goals and principles

Human factors research shows that oversight failures are often driven not by lack of information, but by predictable cognitive effects such as automation bias, complacency, and inappropriate reliance. Decades of work in human–automation interaction demonstrate that humans are poor at sustained monitoring of highly reliable systems and tend to over-trust plausible automated outputs while under-trusting unfamiliar or conflicting recommendations. These findings motivate our emphasis on verification-centred oversight, rather than assuming that transparency or human presence alone improves safety, our design principles aim to structure interactions so that human effort is focused on judgment, contestation, and intervention at points where it meaningfully changes outcomes.

5.1 What good oversight must achieve

Oversight should increase human understanding, control and accountability without collapsing AI into rule-following automation. Building on the solve-verify asymmetry, the design goal is to make verification efficient, targeted, and reliable so humans can judge plausibility, detect errors, and intervene when needed. Three outcomes matter:

- **Control.** Humans can pause, redirect, or substitute decisions, but these interventions are timely and well-targeted, not symbolic or unnecessarily intrusive.
- **Contestability.** Decisions reduce to comprehensible reasons with evidence so they can be reviewed and appealed.

- **Competence.** Oversight improves overall performance and fairness rather than degrading correct AI outputs through blunt overrides.

By accountability we mean the capacity to identify which agent — human or organisational — bears responsibility for a given decision, supported by sufficient records to reconstruct the reasoning and evidence behind it. By contestability we mean not merely the theoretical possibility of challenging a decision, but the practical availability of comprehensible reasons, supporting evidence, and a procedural pathway through which affected parties can raise objections and receive a substantive response.

5.2 Using the four agency conditions to shape oversight

The four conditions outlined earlier, individuality, source of action, goal-directedness, and adaptivity, do not only describe agency but also guide how oversight should be designed.

- **Individuality** requires clear boundaries between the operative role of the AI and the evaluative role of the human. The AI can be free to generate solutions without unnecessary interference, while humans remain positioned to evaluate the results at meaningful checkpoints. When these boundaries are blurred, attribution and responsibility become diffuse, leading to a situation where agency seems to lie everywhere and nowhere. Oversight therefore needs explicit handover points, supported by identity scoping and logging, so that AI retains operative autonomy in producing solutions and humans retain evaluative agency in determining when and how to intervene. *At design time, define the system boundary and human handovers; at runtime, record when control actually crossed that boundary.*
- **Source of action** highlights the need to trace causal provenance in ways that distinguish between generative processes and evaluative decisions. AI systems produce solutions through exploratory and often opaque generative pathways. Humans may feel they exercise control by shaping prompts or iterating instructions, but this is an illusion of control over the generative process itself. This is one reason why U.S. copyright law [47] does not recognise AI-generated content as human-authored, no matter how complicated or labour-intensive the prompting may be. Where human oversight matters are not in co-owning the generative process, but in evaluating outcomes: assessing provenance records of prompts, inputs, AI calls, and interventions, and deciding whether the results should be accepted, modified, or rejected.
- **Goal-directedness** should be understood as a layered structure in which the AI pursues sub-goals while humans retain authority over top-level objectives. AI systems can explore pathways and produce sub-goals that humans cannot or need not fully oversee in real time. The human role is to exercise evaluative judgment over whether these sub-goals align with higher-level aims and constraints such as legal criteria, fairness targets, or policy goals. Making these objectives explicit and traceable allows humans to judiciously evaluate not just the final outcome but also whether the direction of AI exploration remains aligned with legitimate goals. In this way, evaluative agency complements the AI's operative agency without collapsing it into mechanical rule-following. *At design time, publish objectives, constraints, and trade-off rules; at runtime, test whether the current output and sub-goals align with those declarations.*
- **Adaptivity** further clarifies the distinction between operative and evaluative roles. AI systems may adapt at low levels, such as updating internal weights or refining token-level predictions, in ways that humans cannot meaningfully oversee. Attempting to evaluate every micro-adjustment and learning would be infeasible and counterproductive. Human oversight is better targeted at higher-level adaptations that can be expressed in human-understandable terms: changes to policies, constraints, or significant drifts in system behaviour that occur even without explicit re-training. Tools such as versioning, drift detection, and abstain mechanisms allow humans to evaluate and control major shifts while leaving the AI system free to adjust at the micro-level. This division ensures that AI retains adaptive operative capacity, while humans maintain evaluative control over consequential changes. *At design time, set change controls and drift monitors; at runtime, check whether observed behaviour triggers abstain, rollback, or staged rollout criteria.*

These conditions create the scaffolding for non-zero-sum agency. AI systems retain operative agency in performing tasks, while humans, equipped with verification surfaces linked to these four conditions, exercise evaluative agency in deciding when and how to intervene. Tables 3 and 4 align design-time/system-level and run-time/instance-level evaluations to the four agency conditions.

Table 3 System-level evaluation at design time

Agency condition	AI artefacts to expose	Human evaluative checks	System verification surface and sample metrics
Individuality	Architectural boundaries, component map, handover API contracts	Boundary clarity, role separation, identity scopes, auditability	Boundary registry, handover contract tests, % flows with explicit handovers; include a catalogue of model sub-parts eligible for mechanistic probes and their validation status
Source of action	End-to-end provenance model, event schemas, tool integrations	Completeness and integrity of causal capture, time sync, tamper resistance	Provenance schema conformance, log coverage, replayability; document any internal signals admitted as evidence, with calibration tests and limits
Goal-directedness	Policy graph, objective functions, constraint catalogue, trade-off rules	Fitness of goals vs policy, fairness targets, criteria, parameter ranges	Goal ledger with policy links, counterfactual policy tests, fairness sandbox; mapping from goals to explanation requirements and to any validated internal probes tied to goal-relevant behaviours
Adaptivity	Versioning plan, drift monitors, rollout strategy, rollback/abstain design	Change control thresholds, alert quality, red-team playbooks	Change ledger, drift detection precision/recall, rollout failure rate, time-to-rollback; document which drifts trigger external explanations vs internal probe checks

Table 4 Instance-level evaluation at runtime

Agency condition	AI artefacts to expose	Human evaluative checks	Instance verification surface and sample metrics
Individuality	Instance handover records, actor identities, session context	Were boundaries respected and control transferred at the right points	Instance boundary view, handover timestamps, intervention latency; explanation pack indicates whether any internal probe was used
Source of action	Stepwise provenance of prompts, tools, model calls, human edits	Does the causal chain and any admitted internal signal justify the step	Causal trace viewer, missing-evidence alerts, veto precision, acceptance-with-reasons rate; include lightweight saliency or probe readouts only if validated for this class of instance
Goal-directedness	Goals and constraints applied, rationale linked to policy	Do outputs and sub-goals align with declared objectives and acceptable trade-offs	Goals panel with policy links, disagreement analytics, instance-level fairness delta; counterfactuals and contrastive explanations as part of the explanation pack
Adaptivity	Active versions, drift state, abstain/rollback flags	Is behaviour consistent with current versions; do drift or novelty triggers require action	Drift score, abstain utilisation, rollback correctness; show whether drift evaluation relied on external metrics or internal probes

it supports the solve-verify asymmetry so AI keeps solving while humans verify and intervene.

Many of the mechanisms in the catalogue directly respond to known human factors risks. For example, divergence detection and independent checking counter automation bias by introducing friction when systems agree too easily; calibrated confidence and abstention mitigate over-reliance by routing uncertain cases to human judgment; and sampling and post-hoc review address complacency by maintaining reviewer engagement even when systems perform well.

These mechanisms work as paired moves. At design time (system-level evaluation), the organisation commits to a target level of understanding for each angle, for example medium-grained causality through full provenance capture,

6 Oversight mechanism catalogue

Effective oversight needs concrete mechanisms that expose the right artefacts to humans and place control at the points where it adds value. The catalogue below (Table 5) collects practical controls that teams can implement. Each mechanism states what it is, whether it sits at design time or at runtime, which agency dimension it strengthens, and how

Table 5 Oversight mechanisms by layer, agency dimension, and solve-verify role

Mechanism	What it is	Layer	Agency dimension improved	Solve-verify contribution
Boundary registry and handover contracts [21, 48]	Declared system boundary, explicit human–AI handovers, API and role contracts	Design-time, then enforced at run-time	Individuality	Lets AI generate within its zone; lets humans verify boundary crossings and intervene at checkpoints
Identity and session scoping [21]	Stable identities for AI/agents, tools, humans; scoped sessions per case	Design-time and run-time	Individuality	Separates who solved from who verified; enables accountable interventions without touching generation
Provenance capture and causal trace [49, 50]	End-to-end logging of inputs, prompts, tools, model calls, edits	Design-time capability, run-time use	Source of action	Human verifies the decisive step in the chain instead of re-solving the task
Structured rationale schema [27]	Machine- and human-readable reasons aligned to criteria, with fields for evidence and uncertainty	Design-time schema, run-time population	Goal-directedness	Human verifies plausibility and policy fit without reproducing the solution
External explanation pack [51, 52]	Compact bundle with rationale, citations, contrastive views, confidence, abstain flags	Run-time	Source of action; Goal-directedness	Human verifies outcome and reasons at coarse grain when internals are opaque
Validated mechanistic probe hooks [53]	Limited internal signals with documented validity and limits	Design-time adoption, optional run-time use	Source of action; Adaptivity	Enables targeted steerability where evidence supports it; complements external verification
Confidence and uncertainty calibration [54]	Calibrated scores with thresholds tied to impact	Design-time calibration, run-time routing	Source of action	Routes easy cases to light checks and hard cases to deep verification rather than re-solving all cases
Divergence detection and independent checker [21]	Comparison against a second AI/system, heuristic, or ruleset with alerting	Design-time setup, run-time use	Source of action	Flags cases needing verification when solvers disagree; avoids blind trust
Confidence- and impact-aware routing [55, 56]	Triage that considers uncertainty and consequence	Design-time rules, run-time execution	Goal-directedness	Directs human effort where verification changes outcomes
Veto, substitution, approve-with-reasons [57]	Concrete evaluator controls with reason capture	Run-time	Individuality; Goal-directedness	Human verifies and then acts; AI remains the solver until a justified override
Circuit breaker and safe stop	Immediate halt on pre-defined risk triggers	Design-time triggers, run-time action	Adaptivity	Human stops harmful solving without micromanaging generation steps
Goals ledger and policy links [27]	Registry of objectives, constraints, trade-offs, legal hooks	Design-time, referenced at run-time	Goal-directedness	Human verifies alignment of outputs and sub-goals to declared aims
Trade-off disclosure and parameter bounds	Transparent sliders or ranges for cost, risk, fairness	Design-time design, run-time adjustment	Goal-directedness	Human verifies and steers high-level goals without re-engineering the solver
Adaptivity ledger: versions and change control	Version history, rollout plan, rollback, change notes	Design-time governance, run-time status	Adaptivity	Human verifies behaviour against the active version rather than internals
Drift and novelty monitors [58]	Statistical and domain monitors with thresholds and alerts	Design-time selection, run-time review	Adaptivity	Human verifies that solving remains within trained regime; triggers deeper checks
Shadow mode and staged rollout [18, 54]	Parallel running and progressive exposure	Design-time plan, run-time execution	Adaptivity; Source of action	Lets AI solve while humans verify deltas before full substitution
Counterfactual and contrastive probes [20]	What-if questions and minimal-change explanations	Run-time	Source of action; Goal-directedness	Human verifies decision boundaries without re-solving the whole case
Appeal bundle packaging [30, 59]	One-click export of reasons, evidence, versions, and provenance	Design-time format, run-time export	Individuality; Source of action	Supports external verification by reviewers and courts
Sampling for post-hoc review	Risk-weighted case sampling to maintain skill and detect drift	Design-time plan, run-time execution	Adaptivity; Goal-directedness	Human verifies a subset to validate that solving remains reliable over time

or fine-grained change control through validated drift monitors and rollback plans. At runtime (instance-level evaluation), evaluators consume the corresponding instance artefacts: handover records that make boundaries visible,

causal traces that identify the decisive step, goal panels that link outputs to policy and fairness criteria, and drift scores that explain behaviour at the time of generation. This pairing increases predictability and steerability without forcing

humans to re-solve the task. For example, a structured rationale schema provides coarse-to-medium goal understanding at design time by fixing required fields, then yields instance-specific reasons and citations at runtime for quick checking. Validated mechanistic probe hooks offer fine-grained model-level insight where evidence supports them at design time, then inform a small number of high-stakes run-time decisions without encouraging day-to-day micromanagement. Provenance capture and causal traces commit the system to medium-grained causality at design time, then let evaluators locate and judge the decisive step in a case rather than reconstruct the whole pathway. Read this way, the catalogue is a menu for selecting the angle and level of understanding you want, and for deciding where that understanding is guaranteed in assurance and where it is consumed in operation.

To make effective use of the catalogue, select a minimal set that covers all four agency dimensions. At design time, prioritise boundary and provenance capabilities, the goals ledger, and adaptivity governance. At runtime, prioritise the explanation pack, calibrated confidence with abstain, provenance viewing, and evaluator levers. Where validated mechanistic probes exist, add them as optional fine-grain signals. The result is a workflow in which the system continues to solve, while humans verify efficiently and intervene only when it changes outcomes.

7 End-to-end oversight pattern examples

The following are illustrative pattern examples for document evaluation, drawn from recurring oversight challenges. The following four patterns are illustrative configurations derived from clustering recurring oversight mechanisms observed across the analysed cases. They are presented in the context of document evaluation to provide concreteness, but represent analytically distinct allocations of operative and evaluative agency rather than an exhaustive taxonomy. Additional domain-specific variants are possible; the four patterns presented here were selected because they recur across contexts and collectively span the principal oversight objectives identified in the framework.

They capture four common modes of evaluation: review (an artefact judged against subjective criteria), conformance (an artefact tested against objective rules), aggregation (many artefacts synthesised into a coherent whole), and prioritisation (many artefacts analysed and ranked). Each pattern combines a subset of mechanisms into a coherent oversight flow. For each, we identify the problem, list selected mechanisms, and show how design-time system-level evaluation and run-time instance-level evaluation

work together so that AI systems continue solving while humans verify and intervene where it matters.

7.1 Pattern 1 - oversight for criteria-aligned assessment (document A against criteria B)

Context

In many domains, such as research funding, regulatory oversight, or academic publishing, submissions like grant proposals, planning applications, or scholarly papers (A) must be **evaluated against well-defined criteria described in policies, rubrics, or guidelines (B)**.

Problem

As evaluation workloads increase and AI systems become more deeply embedded within these processes, several risks arise that may undermine trust, transparency, and accountability. One central concern is the presence of incomplete or missing evidence to substantiate claims of compliance, which can lead to erroneous or unjustified decisions. Another is the tendency for human reviewers to rubber-stamp outputs that appear plausible, without critically examining the underlying reasoning. This dynamic can result in cases where a decision is technically correct but rests on flawed or incomplete justification, referred as the “right answer, wrong reasons” problem. These risks are amplified at scale, where the complexity and volume of evaluations exceed the capacity of humans to manually track, assess, and verify each step of the process.

Solution

Potential mechanisms

- Structured rationale schema aligned to B (one row per criterion: decision, evidence excerpt, policy clause, uncertainty).
- Goals ledger with links to B and versioning of B.
- Provenance capture and causal trace for inputs, retrieval, tools, and human edits.
- External explanation pack (claim-evidence table, citations to B, contrastive, confidence, abstain flags).
- Constrained retrieval that only cites B (or approved sources).
- Divergence detection against a simple rules/rubric checker.
- Counterfactual/contrastive probes for borderline criteria.
- Calibrated confidence with abstain.
- Risk-weighted sampling and post-decision audit.
- Veto, substitution, approve-with-reasons; appeal bundle packaging.

Translate B into a goals ledger and a mandatory rationale schema so every decision is traceable to a clause and evidence. Configure provenance to capture the decisive steps

and retrieval calls. Calibrate confidence at the criterion-level, and define abstain rules for low evidence or out-of-scope items. Stand up a lightweight rules checker to power divergence alerts. Set audit sampling rates that rise with risk or low confidence. Run shadow evaluations before tightening the rubric.

Runtime instance-level oversight

The system produces a recommendation plus an explanation pack that includes the filled rationale table with citations and a concise causal trace. The reviewer reads the rationale instead of re-doing the assessment, uses contrastive on close calls, and inspects divergence alerts where the rules checker disagrees. A proportion of cases are spot-checked at depth to ensure evidence truly supports each criterion and that reasoning traces are consistent with the cited passages. Reviewer actions (approve, veto, substitute) record reasons and feed the audit queue and appeal bundle.

Solve-verify fit

The AI solves by mapping A to B and assembling evidence-backed reasons; the human verifies claim-evidence links, policy fit, and flagged disagreements, intervening only where it changes outcomes.

Signals to track

Verify time per case; evidence-coverage per criterion; disagreement rate with rules baseline; veto precision; audit reversal rate; incidence of “right answer, wrong reasons”; fairness deltas on protected slices.

Benefits

- **Transparency:** Structured rationales, goals ledgers, and provenance capture make every decision traceable to explicit policy clauses and supporting evidence, enabling clear review and contestation.
 - **Accountability:** Divergence detection, calibrated confidence with abstain, and audit sampling ensure systematic oversight, with human interventions and outcomes fully logged for appeals and governance.
 - **Efficiency:** Explanation packs and constrained retrieval allow humans to focus on verifying claim-evidence links rather than re-performing the entire evaluation process.
 - **Scalability:** Risk-weighted sampling and automation of routine checks enable large-scale evaluation while maintaining oversight quality.
- Drawbacks**
- **Complexity:** Implementing goals ledgers, provenance systems, and divergence detection increases system and governance complexity, requiring specialized expertise.
 - **Cost:** Building and maintaining structured oversight processes, including shadow evaluations and continuous policy versioning, demands significant resources.

- **Verification Burden:** For borderline or novel cases, human reviewers may still need to conduct deep spot-checks, reducing efficiency gains at scale.
 - **Signal Dependence:** The effectiveness of oversight relies on the quality of generated rationales, provenance data, and alerts; poor or noisy signals can mislead reviewers.
- Known uses**
- **AAAI AI-Powered Peer Review System [31]:** AAAI has launched a pilot program that incorporates LLMs to enhance the academic paper review process for the AAAI-26 conference. The LLM generates an additional review alongside traditional human expert evaluations and assists the Senior Program Committee by summarizing reviewer discussions, highlighting key areas of consensus and disagreement among reviewers. All LLM-generated content is thoroughly reviewed by human experts, with comprehensive monitoring and evaluation in place to ensure quality and accountability.
 - **Revelo eReviewer [32]:** eReviewer is an LLM-powered platform designed for grant proposal peer review to support criteria-aligned evaluations. It helps reviewers by automatically summarizing proposals and providing key highlights mapped to predefined rubrics. Human experts oversee the entire process, reviewing all AI-generated outputs and making the final decisions. Detailed audit trails record every evaluation step to ensure transparency and traceability.
 - **Vertesia [33]:** Vertesia leverages LLMs to streamline grant proposal evaluations. The system uses retrieval-augmented LLMs to match proposals with explicit program rubrics, generate initial scores, and produce comprehensive proposal summaries. Human reviewers verify, adjust, or override AI recommendations to ensure decisions are based on expert judgment. Oversight mechanisms include interactive chatbot explanations, clear alignment with program criteria, and full logging of evaluations for auditing and governance.

7.2 Pattern 2 - oversight for conformance checking (document A against manual/guardrail B)

Context

A completed document (A) is typically evaluated against a conformance reference framework (B), such as a style manual, organizational policy or strategy, or a predefined set of guardrails. This evaluation step is an established part of many workflows and is used to ensure that documents meet expected standards. The document could be authored entirely by humans, generated by AI, or created through a combination of both.

Problem

Because B is often lengthy and nuanced, strictly enforcing it during the drafting phase is unreliable. Moreover, the limited controllability of generative systems makes “write-to-spec” approaches difficult to achieve. As a result, a separate checking step is required to assess conformance to B and, where necessary, recommend minimal edits to bring the document into compliance.

Solution

Potential mechanisms

- Goals ledger separating hard constraints (machine-checkable rules) from soft rubrics (judgement-based norms).
- External explanation pack with per-issue rationales, highlight maps, minimal-change diffs, and links to relevant clauses in B.
- Provenance capture for sources, tools, and exemplar sets consulted by the checker.
- Counterfactual rewrites that show the smallest change needed to pass a soft rubric.
- Divergence detection when A’s tone or structure falls outside the exemplar band.
- Calibrated abstain for ambiguous cases requiring human judgement.
- Divergence detection and independent checker: eviwer calibration using a golden set; inter-rater agreement tracking.
- Sampling and post-edit audit; rollback of rubric changes over time.
- Appeal bundle packaging with diffs, rationales, and clause references.*Design-time system-level oversight*

Encode B into a two-layer goals ledger. Make hard constraints machine-checkable with precise tests. Express soft rubrics as evaluable checklists and labelled exemplars; set parameter bounds to balance concision, nuance, and voice preservation. Define the explanation pack format so rationales, highlights, and minimal-change diffs are mandatory. Configure linters and exemplar detectors with thresholds and calibration data; define abstain criteria for low certainty. Train reviewers on a golden set and measure agreement.

Runtime instance-level oversight

The checker returns a conformance report and, where suitable, suggested minimal edits. Hard issues present failing checks with exact locations and clause links. Soft issues present exemplar-linked rationales, highlight maps, and smallest-change diffs so reviewers can accept or reject edits while preserving voice. Provenance identifies any exemplar sets or tools used. Divergence alerts trigger when tone or structure moves outside the exemplar band. Ambiguous cases invoke abstain for human judgement. Reviewers

accept targeted diffs, reject over-corrections, record reasons, and submit cases to sampling or post-edit audit as required.

Solve-verify fit

The AI checks and proposes minimal fixes; the human verifies conformance and decides which edits to adopt. Generation is not steered in real time; conformance is achieved through a clear, auditable checking step.

Signals to track

Hard-check pass rate; agreement with golden set on soft norms; edit acceptance rate by issue type; inter-rater agreement; time to acceptable draft; proportion of abstain cases resolved without rework; rollback frequency after rubric changes; user-reported voice preservation.

Benefits

- **Transparency:** Explanation packs with rationales, highlights, and clause links make every check and suggested edit auditable and easy to trace back to policy or rubric sources.
- **Consistency:** Goals ledgers and calibrated linters standardize the application of hard rules and soft rubrics, reducing reviewer subjectivity and drift over time.
- **Efficiency:** Minimal-change diffs and calibrated abstain thresholds focus human effort on ambiguous cases, improving throughput while preserving author voice.
- **Accountability:** Provenance capture, rollback tracking, and audit sampling provide a verifiable record of checks, tools, and rubric updates for governance purposes.**Drawbacks**

- **Complexity:** Encoding nuanced rubrics into exemplar-based systems and maintaining calibration data require significant design-time effort and specialized expertise.
- **Cost:** Building and sustaining rule linters, exemplar detectors, and audit pipelines demands ongoing investment in tooling and reviewer training.
- **Signal Dependence:** Oversight effectiveness relies on high-quality exemplars, thresholds, and provenance data; poor or biased inputs can mislead both the system and reviewers.
- **Verification Burden:** In borderline or novel cases, reviewers may still need to invest substantial time resolving abstains and contested edits, limiting scalability gains.**Known uses**

- **ValidMind Document Checker [34]:** ValidMind’s LLM-powered Document Checker evaluates model documentation for compliance with regulatory standards such as model risk management policies and AI governance frameworks. The system automatically maps sections of the document to regulatory criteria, identifies gaps, and provides rationales with source evidence.

Human validators remain responsible for interpreting nuanced requirements, confirming or dismissing flagged issues, and finalizing compliance assessments. Every interaction is logged, creating a traceable record of human decisions and AI contributions.

- **Norm AI [35]:** Norm AI provides AI agents that act as an automated compliance reviewer for documents and proposals. These agents analyse content against complex regulations and flag any sections that may violate specific rules or laws, pointing reviewers to the exact regulation clauses at issue.
- **Acrolinx [36]:** Acrolinx's enterprise content governance tool uses AI to scan and score documents against an organization's style guides and compliance standards. It checks writing for clarity, tone, terminology, and adherence to rules, ensuring each document follows the required style and policy guidelines. The system provides suggestions or automated rewrites for flagged issues and links each suggestion to the relevant guideline, helping authors fix content while preserving the intended voice.

7.3 Pattern 3 - oversight for evidence-grounded synthesis and thematic analysis (from large corpus to report)

Context

In many domains, there is a need to synthesize large collections of documents, such as consultation submissions, research papers, or policy reports, into a single structured output. This synthesis typically includes identifying common themes, highlighting nuances and contrasts, and presenting representative quotes along with supporting evidence. The resulting report often serves as a key input for decision-making, strategy development, or policy formation, making accuracy and comprehensiveness essential.

Problem

When synthesizing hundreds of source documents, several risks arise. Important material may never surface, leading to gaps in coverage. Hallucinated or unsourced claims can undermine the credibility of the synthesis, especially when outputs are produced by or assisted with generative systems. In addition, there is a danger of over-confident generalizations that obscure or ignore minority views, resulting in a misleading or biased representation of the underlying evidence. These challenges threaten both the quality and trustworthiness of the final report.

Solution

Potential mechanisms

- Provenance capture for every claim: passage-level anchors and permalinks.
- Structured rationale schema for synthesis: per-theme claim-evidence tables with source counts, stakeholder types, and date ranges.
- Trade-off disclosure and parameter bounds: transparent weighting scheme (incl. stakeholder weights) and publish bounds.
- Divergence detection: independent model or heuristic re-summaries; minority-view mining.
- Counterfactual and contrastive probes: contrastive theme extraction and counter-evidence prompts.
- External explanation pack: quote bundles, source lists, confidence, abstain/low-coverage flags.
- Calibrated confidence and abstain criteria tied to evidence density and source diversity.
- Post-synthesis audit: risk-weighted spot checks; reversal tracking; appeal bundle with all citations. *Design-time system-level oversight*

Define the corpus register with frozen inclusion criteria, metadata normalisation, and dedup rules so the boundary of "what can be said" is auditable. Map the report outline to a synthesis rationale schema: each theme must include a concise claim, supporting passages with anchors, source diversity stats, stakeholder coverage, recency window, and explicit counter-evidence. Configure coverage probes: a checklist of sentinel items (expected stakeholders, landmark papers, seminal arguments) that the system must surface or abstain on. Set weighting rules for stakeholder categories and document quality signals; publish them in a goals ledger. Stand up divergence detection via an independent summariser or rule-based sampler to challenge the main synthesis. Specify the explanation pack format and the audit sampling plan that oversamples low-coverage or high-impact themes.

Runtime instance-level oversight

The system returns a draft report plus an explanation pack. For each theme it shows the claim, representative quotes with page-level anchors, a source list with counts and dates, coverage indicators (e.g., "6 of 8 stakeholder types represented"), and counter-evidence. Abstain is triggered where evidence density or diversity is below thresholds. Reviewers verify by checking anchors, running contrastive prompts on contentious themes, and inspecting divergence alerts where the independent summariser disagrees. Spot checks target sentinel items and minority-view capture. Reviewer actions include accept with reasons, revise scope/weights, request targeted re-search, or mark for audit.

Solve-verify fit

The AI solves by organising themes and assembling verifiable evidence; humans verify coverage and balance,

test disagreements, and intervene where it changes the conclusions.

Signals to track

Evidence-to-claim ratio; source diversity per theme; sentinel-hit rate; minority-view capture rate; abstain utilisation; audit reversal rate; quote anchor validity; recency distribution; time-to-verify per theme.

Benefits

- **Transparency:** Provenance capture with passage-level anchors and quote bundles ensures every synthesized claim can be traced back to its original source, improving auditability and trust.
- **Coverage:** Sentinel-based checks, stratified sampling, and diversity metrics reduce the risk of missing important documents, stakeholders, or minority viewpoints.
- **Balance:** Weighting schemes and divergence detection prevent over-confident generalizations by highlighting conflicting perspectives and surfacing counter-evidence.
- **Accountability:** Structured rationale schemas, audit sampling, and appeal bundles create a clear, verifiable trail of how conclusions were reached and validated.
- **Scalability:** Automating theme extraction, evidence assembly, and coverage probes allows large-scale synthesis while maintaining systematic human oversight.

Drawbacks

- **Complexity:** Configuring registers, coverage probes, weighting rules, and divergence detection requires significant design-time expertise and ongoing calibration.
- **Cost:** Building and maintaining provenance infrastructure, explanation packs, and audit pipelines demands sustained resources and skilled reviewers.
- **Verification Burden:** Contentious or ambiguous themes may still require intensive manual checking, especially when abstain flags or divergence alerts are frequent.
- **Signal Dependence:** Oversight effectiveness relies on the accuracy of anchors, thresholds, and sentinel definitions; weak or biased inputs can lead to misleading synthesis.
- **Governance Load:** Managing versioning, inclusion criteria, and rollback processes for evolving corpora and policies introduces operational overhead.

Known Uses

- **Consensus [37]:** Consensus is an LLM-based literature search tool that scans a large corpus of 200+ million papers and produces a concise synthesis of findings, with citations for every claim. For each query, it retrieves top relevant papers and uses GPT-based summarisation to answer the question while tracing each insight back to its source.

- **Elicit [38]:** Elicit assists researchers in synthesising findings from academic papers using AI. It can retrieve relevant studies for a given question and extract key points or quotes from each, effectively building an evidence table. It prioritises grounding outputs in the original documents, enabling users to trace claims back to the literature.
- **Scite Assistant [39]:** Scite is a scholarly assistant that leverages a large database of scientific publications and an LLM to answer questions with evidence. It uses an AI model (GPT-based) to read through papers and produce an answer with source-based grounding. Scite's system distinguishes supportive vs. conflicting evidence by analysing how each cited paper discusses a topic, thus highlighting reliable research. The tool's output includes citations to the papers it read, helping users verify claims and assess evidence quality.

7.4 Pattern 4 - oversight for signal detection, novelty, and prioritisation across streams (from news/reports to ranked issue list)

Context

In many operational settings, organizations must continuously scan high-volume information streams, such as media reports, public advisories, and internal incident reports, to identify emerging issues that warrant attention. These streams are monitored against explicit criteria to determine which items are relevant and to prioritize them for subsequent action. This process supports timely decision-making and effective allocation of resources in dynamic environments.

Problem

Several challenges arise in managing this type of monitoring and prioritization workflow. Important signals may be missed entirely, resulting in low recall and the potential for delayed or inadequate responses. Conversely, overly sensitive detection can lead to alert fatigue, where large volumes of false positives overwhelm reviewers and erode trust in the system. In addition, rankings may be shallow or opaque, providing little insight into why certain issues are prioritized, which hinders accountability and informed decision-making. Over time, concept drift, shifts in the meaning of signals or criteria, can degrade system performance, requiring ongoing recalibration and oversight.

Solution

Potential mechanisms

- **Trade-off disclosure and parameter bounds:** criteria-to-score mapping with transparent weights and adjustable bounds.

- Provenance capture and causal trace: per-candidate evidence vouchers (snippets, links, timestamps).
- Drift and novelty monitors: "first-seen" novelty detectors; similarity clustering and collapse.
- Divergence detection via second model or heuristic rules.
- External explanation pack per issue: score breakdown, supporting evidence, known gaps, and confidence.
- Confidence- and impact-aware routing; abstain for low evidence or low novelty.
- Sampling for post-hoc review: risk-weighted post-decision sampling; appeal-bundle packaging for escalations.*Design-time system-level oversight*

Register streams and enforce freshness windows and dedup logic so scanning remains predictable. Translate "worthy" into a criteria-to-score function with weights for impact, relevance, novelty, and credibility; publish bounds in the goals ledger. Configure novelty detectors and clustering to collapse duplicates. Define coverage monitors: watchlists and sentinels for topics you must not miss. Calibrate divergence detection and abstain thresholds tied to evidence quality. Specify the per-issue explanation pack (score breakdown, evidence vouchers, gaps, and confidence). Set sampling rates by tier to audit both false positives and false negatives.

Runtime instance-level oversight

The system outputs a ranked list of issues. Each entry carries an explanation pack: score components, supporting snippets with links and times, similarity cluster context, novelty flags, and any watchlist hits or misses. Items with low evidence or novelty trigger abstain for human review. Reviewers verify top-ranked items first, check provenance and cluster context, and use counter-example mining to reduce spurious spikes. Disagreements with the secondary checker are investigated. Reviewers can reweight criteria within bounds, escalate, or dismiss with reasons; dismissed items feed the learning set for future precision.

Solve-verify fit

The AI solves by detecting and ranking candidate signals; humans verify the ranking rationale, adjust within transparent bounds, and focus attention where it changes downstream action.

Signals to track

Precision/recall against adjudicated samples; watchlist-miss rate; duplicate collapse rate; reviewer reweight frequency and range; time-to-decision by tier; escalation and reversal rates; drift in score distributions over time.

Benefits

- **Transparency:** Explanation packs with score breakdowns, provenance vouchers, and evidence links make

rankings auditable and justify why each issue was surfaced or prioritized.

- **Precision and Recall:** Counter-example mining, watchlist sentinels, and novelty detection reduce missed signals while minimizing false positives and alert fatigue.
- **Adaptability:** Adjustable scoring weights and divergence detection allow the system to evolve with shifting criteria and changing data sources.
- **Accountability:** Sampling, escalation bundles, and audit trails ensure that both detections and dismissals are reviewable, supporting governance and trust.**Drawbacks**

- **Complexity:** Configuring scoring logic, novelty detectors, clustering, and divergence rules requires specialized expertise and careful ongoing calibration.
- **Cost:** Maintaining registries, provenance tracking, and post-decision audit infrastructure demands sustained resources and operational effort.
- **Verification Burden:** Ambiguous or novel cases may still require intensive manual review, especially when abstains or disagreements are frequent.
- **Signal Dependence:** Performance relies heavily on the quality and stability of streams, watchlists, and thresholds; weak or drifting signals can undermine accuracy over time.**Known Uses**

- **Primer Command [40]:** is an AI-driven situational awareness platform that monitors over 60,000 global news and social sources in real time. It uses LLM to eliminate duplicates, extract entities (people, places, organizations), and provide structured event updates with "human-grade" precision. The system separates signal from noise, flagging emerging developments or misinformation spikes and presents them on a single dashboard. Primer Command offers explainable insights and allows human operators to apply filters or geofences, ensuring critical events are surfaced for review while less relevant chatter is filtered out.
- **Dataminr [41]:** provides a real-time AI platform that discovers the earliest signals of high-impact events, risks, and threats from public information streams. It continuously scans over a million public data sources worldwide, including social media, news sites, blogs, sensors, to detect anomalies and breaking events faster than any other source. Human analysts can review these AI-generated "briefs," trace the underlying sources, and decide on responses.
- **Signal AI [42]:** offers an external intelligence platform for risk and reputation management, powered by its proprietary AI engine called AIQ. The system ingests 5+ million news articles, regulatory filings, and social media posts per day to "find the signal in the noise" for

each client. A discriminative model first retrieves the most relevant data points, then a generative model produces concise insights which are grounded in the source material to minimise hallucinations. Humans supervise the process by tuning alert criteria, validating AI findings, and relying on audit trails that link back to original sources.

8 Limitations and challenges

While the proposed framework aims to strengthen human oversight by reallocating agency between AI systems and human evaluators, it also introduces important trade-offs and limitations. This section critically examines these issues to clarify the scope, feasibility, and boundary conditions of the approach.

8.1 Scalability of evaluative agency

A central limitation of the framework concerns the scalability of evaluative agency. By design, the framework shifts oversight away from passive monitoring toward active verification, contestation, and judgment. This may effectively **create a distinct class of labour-human evaluators responsible for reviewing explanation packs, provenance records, and justification artefacts**. Such work entails non-trivial training requirements, sustained cognitive effort, and institutional support, particularly in complex or high-volume decision environments.

Importantly, this burden should be understood as a redistribution rather than a net creation of effort. In many high-stakes public-sector contexts, similar verification work already occurs informally, inconsistently, or retrospectively, often without adequate tooling or accountability. The framework makes this labour explicit, structured, and auditable, but does not eliminate its cost. As a result, the feasibility of evaluative agency depends on organisational capacity, case volume, and the availability of trained personnel. In settings where such capacity is limited, evaluative agency may need to be selectively applied or supplemented by sampling and escalation strategies.

8.2 The “rubber stamp” problem revisited

Although the framework is explicitly designed to avoid superficial human-in-the-loop oversight, it does not eliminate the risk of automation bias altogether. In particular, well-structured explanation packs may themselves become a new locus of undue trust. Coherent, authoritative, or consistently formatted rationales risk being accepted at face

value, potentially shifting automation bias from model outputs to model justifications.

This limitation highlights that **explanation alone is not a sufficient safeguard**. Within the proposed framework, explanation artefacts are treated as inputs to verification rather than as evidence of correctness. Mechanisms such as divergence detection, comparative evaluation, sampling, and contestation triggers are intended to reintroduce friction and signal when explanations warrant closer scrutiny. Nonetheless, the effectiveness of these mechanisms depends on organisational norms, reviewer incentives, and the extent to which evaluative practices are reinforced rather than treated as procedural formalities.

8.3 Economic and organisational feasibility

Several mechanisms proposed in the framework, such as comprehensive provenance capture, persistent goals ledgers, and fine-grained justification logging are non-trivial to implement in the real-world context. They **require engineering effort, integration with existing systems, and organisational commitment** to governance processes that may not be uniformly present across institutions.

Consequently, the framework should not be interpreted as a universal baseline applicable to all AI deployments. Instead, it is best understood as a reference design whose components can be selectively adopted based on risk profile, organisational maturity, and regulatory context. For lower-risk or resource-constrained settings, partial adoption may be more realistic, focusing on a subset of mechanisms that deliver the greatest governance value relative to implementation cost.

8.4 Boundary cases and infeasible contexts

The layered agency model assumes that evaluative agency can be exercised with sufficient time, contextual information, and institutional authority to support deliberation and contestation. This assumption does not hold in all domains. **In real-time, safety-critical settings (e.g., autonomous driving, collision avoidance systems, or high-frequency trading) human evaluative intervention at run time may be impracticable or unsafe.**

In such cases, oversight must shift toward design-time assurance, simulation, stress testing, and post-hoc accountability rather than real-time evaluation. The framework is therefore not intended to replace existing safety engineering approaches in these domains, but to complement them in contexts where human judgment can be meaningfully exercised. Explicitly recognising these boundary conditions is essential to avoiding misapplication of the framework.

8.5 Limitations of the known-use analysis

The known-use analysis relies primarily on publicly available documentation, including vendor technical materials, whitepapers, and conference reports. Such sources are often marketing-oriented and may emphasise intended safeguards rather than operational limitations or failure cases. As a result, the mechanisms identified in Table 2 reflect documented design affordances rather than independently verified practices. In particular, public materials may underreport breakdowns in oversight, organisational workarounds, or incentive misalignments that materially affect how oversight functions in practice.

This limitation constrains the evidentiary status of the analysis. The purpose of the known-use study is therefore not to evaluate the effectiveness of these mechanisms, but to identify recurring structural features that inform the abstraction of the mechanism catalogue (Table 5). Empirical validation (e.g., field studies, audits, or controlled experiments) remains necessary to assess how these mechanisms operate under real organisational conditions and to examine potential gaps between documented design and lived practice.

9 Conclusion

Human oversight in AI systems is often framed as a trade-off between human control and machine autonomy. This paper has argued that such a framing is misleading. Rather than treating oversight as a zero-sum constraint on AI capability, we propose a layered conception of agency in which AI operative agency and human evaluative agency are structured to coexist. By distinguishing these forms of agency and translating them into architectural design constraints, the framework provides a principled alternative to both over-constraining automation and symbolic human-in-the-loop oversight.

The paper makes three central contributions. First, it introduces a layered agency model grounded in a frame-dependent account of agency, clarifying where and how accountability attaches in AI-enabled decision processes. Second, it reframes the explainability requirement by arguing that external reasoning faithfulness can be sufficient, and in many cases preferable, for enabling meaningful oversight. Third, it presents a structured catalogue of oversight mechanisms and four end-to-end design patterns that operationalise these principles across the decision stack. Together, these contributions move the discussion of oversight from abstract normative aspiration toward implementable system design.

At the same time, the framework remains conceptual and design-oriented. Its effectiveness depends on organisational

capacity, human training, and the quality of institutional incentives that sustain evaluative agency. Future work should therefore pursue empirical validation across domains. This includes controlled studies examining whether explanation packs and divergence detection mechanisms reduce inappropriate reliance, field deployments testing the scalability of evaluative roles in public-sector contexts, and comparative evaluations of internal versus external reasoning faithfulness in supporting justified trust. Longitudinal studies are particularly important to assess whether structured oversight mechanisms mitigate automation bias over time or introduce new forms of routinised compliance.

Beyond empirical testing, further research is needed to refine boundary conditions for the layered agency model. Real-time safety-critical domains may require alternative assurance strategies centred on design-time verification rather than run-time evaluation. Additionally, economic and organisational feasibility constraints must be better understood to determine which mechanisms provide the highest governance value relative to implementation cost. Finally, integrating the framework with existing safety-case methodologies and regulatory regimes presents an opportunity to bridge conceptual oversight models with operational compliance practices.

Human oversight is not achieved by inserting a human into the loop, but by designing systems in which agency, accountability, and evaluative capacity are structurally embedded. By articulating agency as an architectural constraint and providing implementable design patterns, this work aims to contribute a concrete foundation for meaningful and sustainable oversight in agentic AI systems.

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Declarations

Ethical Approval This study did not involve human participants, animals, or sensitive data. Ethics approval was therefore not required.

Conflict of interest The authors declare no conflict of interest.

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